Econometrics Chapter 6: Least Squares

(1) Weighted Least Squares Model

1. Model of WLS

$\hat{\beta}_1^{wls} = \frac{1}{n}\sum \frac{Y_i}{X_{i1}}$

- unbiased $E(\hat{\beta}_1^{wls}) = \beta_1$
- linear estimator since $\hat{\beta}_1^{wls} = \sum (\frac{1}{nX_{i1}}) Y_i$

- idea: to weight each observation so that weighted noise terms are no longer heteroskedastic

- Regression Equation:

$$\frac{Y_i}{\sqrt{n_i}} = \beta_0 \frac{1}{\sqrt{n_i}} + \beta_1 \frac{X_{i1}}{\sqrt{n_i}} + \ldots + \beta_{k-1} \frac{X_{ik-1}}{\sqrt{n_i}} + \frac{\varepsilon_i}{\sqrt{n_i}}$$

which can be written as:

$$Y_i^* = \beta_0 X_{0,i}^* + \beta_1 X_{i1}^* + \ldots + \beta_{k-1} X_{i,k-1}^* + \varepsilon_i^*$$

Variance

$$-Var(\hat{\beta}_1^{wls} \mid X_{11}, \ldots, X_{n1}) = \frac{1}{n^2} Var(\frac{Y_i}{X_{i1}} \mid X_{11}, \ldots, X_{n1})$$

$$= \frac{1}{n^2}(n\sigma^2) \text{ since } \frac{Y_i}{X_{i1}} \text{ is iid, remaing heteroskedasticity}$$

$$= \frac{\sigma^2}{n}$$

- comparing GLS & OLS

$$Var(\hat{\beta}_1^{ols} \mid X_{11}, \ldots, X_{n1}) = Var(\frac{\sum X_{i1} Y_i}{\sum X_{i1}^2} \mid X_{11}, \ldots, X_{n1})$$

under Heteroskedasticity

$$= \frac{\sigma^2 \sum X_{i1}^4}{(\sum X_{i1}^2)^2} \geq \frac{\sigma^2}{n}$$

∴ therefore, OLS is inefficient when there is conditional heteroskedasticity

GLS assigns less weight to noisier observations

Heteroskedatic Error

- $E(\varepsilon_i^2 | X) = \sigma_i^2 = \sigma^2 \eta_i \quad , i = 1, \ldots, n$

whereby, we assume η_i is a completely known function of the regressors

  - only unknown is σ^2

- η_i is fixed conditional on the regressors

  - $E(\varepsilon_i^* | X) = E\left(\left(\frac{\varepsilon_i}{\sqrt{\eta_i}}\right) | X\right)$

    $= \frac{1}{\sqrt{\eta_i}} E(\varepsilon_i | X) = 0$

  - $E(\varepsilon_i^{*2} | X) = E\left(\left(\frac{\varepsilon_i}{\sqrt{\eta_i}}\right)^2 | X\right) = \frac{1}{\eta_i} E(\varepsilon_i^2 | X)$

    $= \frac{1}{\eta_i} \sigma^2 \eta_i$

    $= \sigma^2$

  - $E(\varepsilon_i^* \varepsilon_j^* | X) = E\left(\left(\frac{\varepsilon_i \varepsilon_j}{\sqrt{\eta_i}\sqrt{\eta_i}}\right) | X\right) = \frac{1}{\sqrt{\eta_i}\sqrt{\eta_i}} E(\varepsilon_i \varepsilon_j | X) = 0$

Model for Heteroskedasticity (Possible model)

$\eta_i = \exp(\alpha_1 X_{i1} + \ldots + \alpha_{k-1} X_{i,k-1})$

exp ensure variance is +ve

OLS argument for WLS

- $\underset{\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_{k-1}}{\arg\min} \sum (Y_i^* - \hat{\beta}_0 X_{i0}^* - \hat{\beta}_1 X_{i1}^* - \ldots - \hat{\beta}_{k-1} X_{ik-1}^*)^2$

$= \sum \left(\frac{Y_i}{\sqrt{\eta_i}} - \hat{\beta}_0 \left(\frac{1}{\sqrt{\eta_i}}\right) - \hat{\beta}_1 \frac{X_{i1}^*}{\sqrt{\eta_i}} - \ldots - \hat{\beta}_{k-1} \frac{X_{ik-1}^*}{\sqrt{\eta_i}}\right)^2$

$= \sum \frac{1}{\eta_i} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{i1}^* - \ldots - \hat{\beta}_{k-1} X_{ik-1}^*)^2$

$= \sum \omega_i \hat{\varepsilon}_i^2$ whereby, it can be seen as minimizing a sum of weighted square residuals

$\omega_i = \frac{1}{\eta_i}$

Fitted Values

- $\hat{Y}_i^{wls} = \hat{\beta}_0^{wls} + \hat{\beta}_1^{wls} X_1 + \ldots + \hat{\beta}_{k-1}^{wls} X_{k-1}$
- $\hat{\varepsilon}_{i,wls} = Y_i - \hat{Y}_i^{wls}$

$= Y_i - \hat{\beta}_0^{wls} - \hat{\beta}_1^{wls} X_{i1} - \ldots - \hat{\beta}_{k-1}^{wls} X_{ik-1}$

Goodness of Fit

- $R^2_{wls} = 1 - \frac{\sum \hat{\varepsilon}_{i,wls}^2}{\sum (Y_i - \bar{Y})^2}$

  - generally less than R^2 from OLS — less weight on outliers worsen fit
  - can also be -ve as $\sum \hat{\varepsilon}_i^2{}_{wls}$ can be greater than $\sum (Y_i - \bar{Y})^2$

Weighted R^2

- better and prevent negative values
- $R^2_\omega = \frac{\sum \omega_i (\hat{Y}_i - \bar{Y}_\omega)^2}{\sum \omega_i (Y_i - \bar{Y}_\omega)^2}$

where $\bar{Y}_{wls}$ is the weighted mean of $\{Y_i\}_{i=1}^n$